Lecture 11
Blade modal analysis

ASDA/Dynamics of Rotor Systems

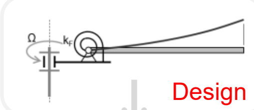
Brano Titurus

QB 2.64 UoB

This lecture

- BVP problem specification in bvp4c
- Example bvp4c implementation in MATLAB
- Parametric explorations in modal analysis of a blade
  - root stiffness and rotor speed effects
  - ω and Ω parameter sweeps

Dynamic design loop in BVP solver



Consider the dynamic design of a rotating flapping blade with the hinge spring design parameter

Static problem

$$\left(EI_F w''\right)'' - \left(T w'\right)' = L$$

$$\begin{bmatrix} w(e_F R) \\ w''(e_F R) - (k_F / EI_F) w'(e_F R) \\ w''(R) \\ w'''(R) \end{bmatrix} = \mathbf{0}$$

Modelling

Harmonic response problem

$$\left(EI_F W''\right)'' - \left(T W'\right)' - \omega^2 m W = L_0$$

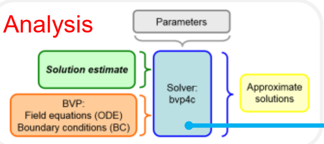
$$\begin{bmatrix} W(e_F R) \\ W''(e_F R) - (k_F / EI_F) W'(e_F R) \\ W''(R) \\ W'''(R) \end{bmatrix} = \mathbf{0}$$

Eigenvalue problem (*modal analysis*)

$$\left(EI_F W_j''\right)'' - \left(T W_j'\right)' - \omega_j^2 m W_j = 0$$

$$\begin{bmatrix} W(e_F R) \\ W''(e_F R) - (k_F / EI_F) W'(e_F R) \\ W''(R) \\ W'''(R) \\ W(R) - 1 \end{bmatrix} = \mathbf{0}$$

Normalization condition (discussed later)



BVP models transformed to the 1st order form

Design loop

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Transcription to the first order form for bvp4c

All field equations need to be written as a set of the first order ODEs.

Consider previously introduced *flapping blade eigenvalue problem*:

$$EI_F W^{iv} - 0.5\Omega^2 m (R^2 - x^2) W'' + \Omega^2 m x W' - \omega_F^2 m W = 0$$

We can use following substitutions: $p = W', q = W'', r = W'''$

Then, the original equation can be written (note the presence of p, q, r):

$$EI_F r' - 0.5\Omega^2 m (R^2 - x^2) q + \Omega^2 m x p - \omega_F^2 m W = 0$$

This equation is now combined with the original substitution identities:

$$W' = p$$
$$p' = q$$
$$q' = r$$
$$r' = \left(0.5\Omega^2 m (R^2 - x^2) q - \Omega^2 m x p + \omega_F^2 m W\right) / EI_F$$

This is the original rotating flapping blade ODE expressed in the 1st order form - this problem formulation is expected by bvp4c function.

The solved for unknown functions: $[W, p, q, r] = [W, W', W'', W''']$

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Summary of the bvp4c problem formulation

Complete problem formulation in bvp4c solver:

$$\frac{d}{dx}\begin{bmatrix} W \\ p \\ q \\ r \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \\ \left(0.5\Omega^2 m(R^2-x^2)q - \Omega^2 m x p + \omega_F^2 m W\right)/EI_F \end{bmatrix}$$

ODEs (W,p,q,r = unknown functions)

$$\begin{bmatrix} W(e_F R) \\ q(e_F R) - (k_F/EI_F)\,p(e_F R) \\ q(R) \\ r(R) \\ W(R)-1 \end{bmatrix} = \mathbf{0}$$

BCs + normalization cond.

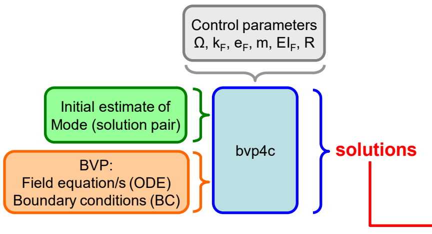
General remarks:

- *bvp4c* requires the *initial solution guess*.
- Each BVP function call attempts to determine one solution (e.g., in case of eigenvalue analysis, one solution pair $\omega_{F,j}$ and $W_j(x)$ is sought).
- *bvp4c* calculates $W_j(x)$ and its higher derivatives.

Normalization condition:

This eigenvalue problem is represented by four field equations, four unknown functions (W,p,q,r) *and* one unknown parameter ω_F. To be able to solve this eigenvalue problem, one additional normalization equation must be formulated. This condition effectively "fixes" otherwise indeterminate magnitude of the sought eigenfunctions (mode shape functions).

Annotated eigenvalue BVP example




```
% Problem parameters
m=10;                        % [kg/m], mass density
EIf=5000; kf= ...            % [kg*m^2], flapping stiffness
l_guess=omega(m,EIf,R,...)^2;
% solution estimate
solinit = bvpinit(linspace(efR,R,Nest),@fcn_init,l_guess);
% solution parameters
options = bvpset('stats','off','RelTol',1e-4);
% field equation in bvp4c format
field_eq=@fcn_ode;
% boundary conditions in bvp4c format
boundary_cond=@fcn_bc;

% solver execution
sol=bvp4c(field_eq,boundary_cond,solinit,options);
ws=sqrt(sol.parameters);
xbld=sol.x; Wi=sol.y(1,:); dWi=sol.y(2,:); ...
figure, grid on, box on, hold on, plot(xbld,Wi)
```

Field equation (ODE):

$$\frac{d}{dx}\begin{bmatrix} w_j \\ p \\ q \\ r \end{bmatrix} = \begin{bmatrix} p \\ q \\ r \\ \left(\Omega^2 m(R^2-x^2)q/2 - \Omega^2 m x p + \omega_{F,j}^2 m w_j\right)/EI_F \end{bmatrix}$$

Boundary conditions:

Normalization condition:

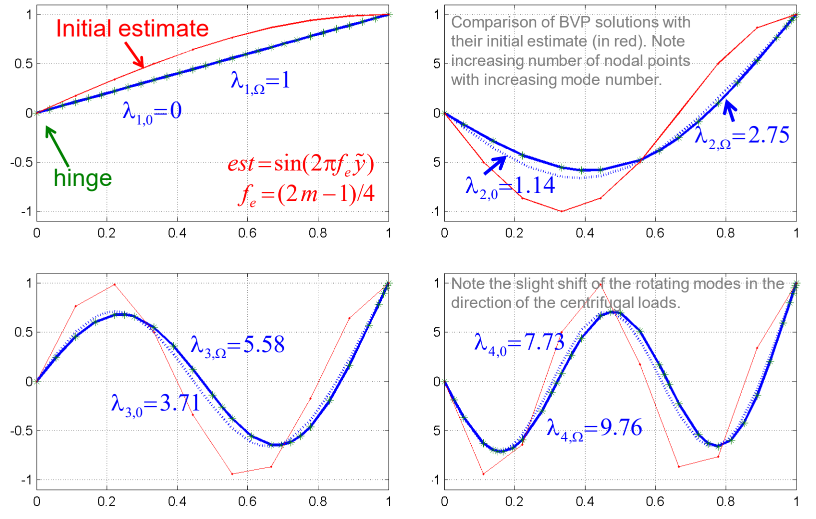
$$\begin{bmatrix} W(e_F R) \\ q(e_F R) - (k_F/EI_F)\,p(e_F R) \\ q(R) \\ r(R) \\ W(R)-1 \end{bmatrix} = \mathbf{0}$$

(tip deflection is equal to 1)

```
function dydx=fcn_ode(x,y,lambda)
   % rotating blade model
   z=y(1);  % shape
   u=y(2);  % diff shape
   v=y(3);  % diff diff shape
   w=y(4);  % diff diff diff shape
   dydx = [ u ; v ; w ; ...
     (lambda*m*z+Omega^2*m*((R^2-x^2)*v/2-x*u))/EIf];
end
function res=fcn_bc(ya,yb,lambda)
   % left       right                BC
   za=ya(1);   zb=yb(1);        %  displacement
   ua=ya(2);   ub=yb(2);        %  D_displacement
   va=ya(3);   vb=yb(3);        %  DD_displacement
   wa=ya(4);   wb=yb(4);        %  DDD_displacement
   res = [ za ; va-(kf/EIf)*ua ; vb ; wb ; zb - 1 ];
end
function v=fcn_init(x)
   % initial guess (shape)
   v = [ ... ];
end
```

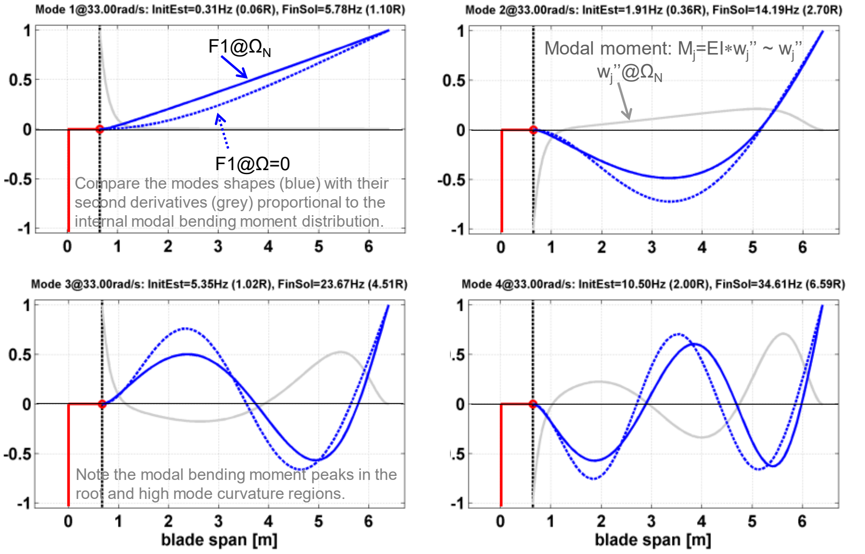
Example of mode shape solutions

The general form of the first four mode shapes of the flapping blade in the non-rotating and rotating states.



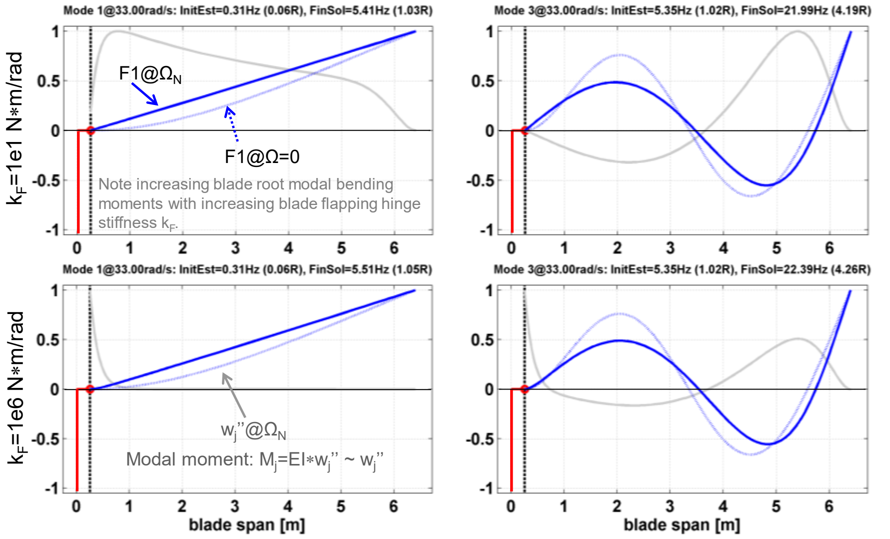
Example of mode shape solutions, cont.

Rotating flapping modes F1, F2, F3, F4 for the configuration with a rigid blade root offset e_F=0.1.



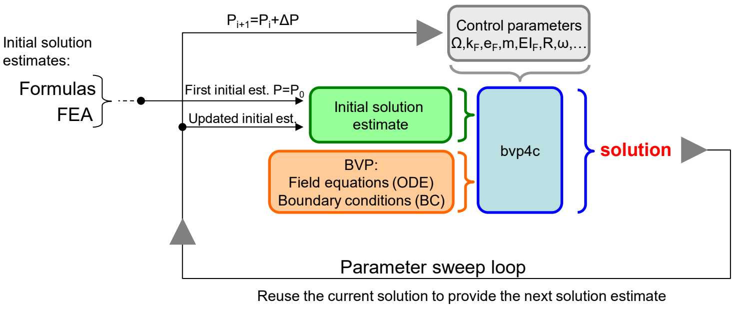
Example of mode shape solutions, cont.

Rotating flapping modes F1 and F3 for configurations with soft and stiff blade root offsets e_F=0.04.



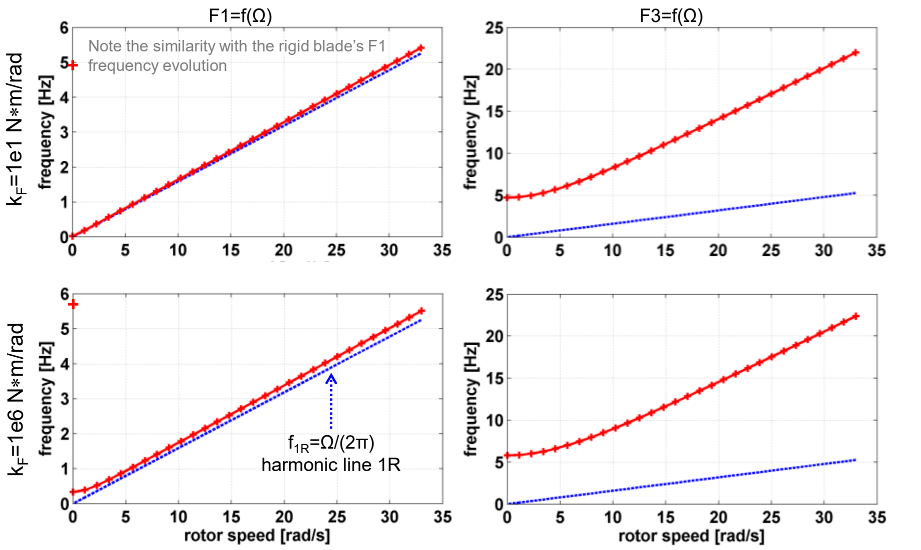
Parametric explorations using bvp4c

BVP solvers (e.g., bvp4c) can be repeatedly called in the programmatic loops to systematically study blade dynamics changes under single and multi-parameter variations. This approach can be used to perform design and optimisation studies. This strategy also enables computation of “difficult-to-reach solutions” using direct parameter continuation. In our context, two typical candidate parameters are the excitation frequency ω in harmonic analyses and the rotor speed Ω in eigenvalue analyses.



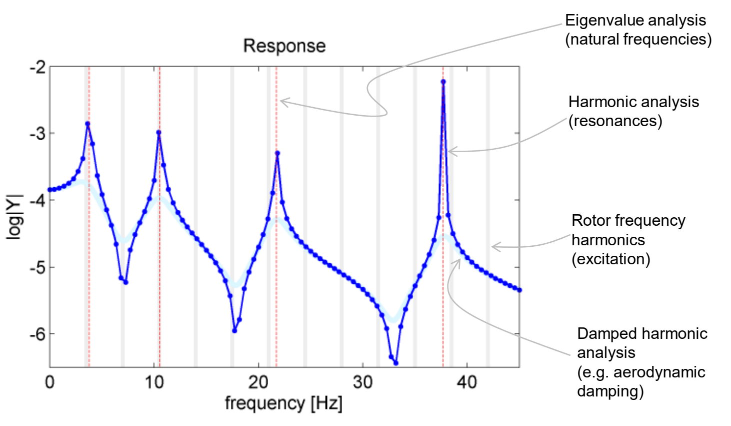
Parametric study: frequency diagrams

Plotting the loci of the (real) eigenvalues of the first four rotating flapping modes as a function of the rotor speed to construct the frequency diagram.



Parametric study: Frequency Response Functions

Consider the previously introduced (annotated) bvp4c example:



These results were generated for $F(t)|_{x=R}=F_0e^{i\omega t}$, where F_0=1. The resulting harmonic response (blue) represents the function $h(\omega)=Y_{tip}/F_0$, where $h(\omega)$ is the Frequency Response Function of the rotating flapping blade.

Summary

- bvp4c solver requires conversion of the original BVP to the first order ODE form.
- Suitable normalisation conditions need to be included in the problem formulation when solving eigenvalue BVPs.
- bvp4c solver determines the unknown deflections (or amplitudes) as well as their higher order derivatives.
- Repeated BVP calls are used to perform parametric studies and construct important design diagrams.